Project Notes

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Project title: Small-time controllability of control systems on 3D Lie groups

Project description: The goal of the project is to determine small-time controllability of affine control systems with one control on 3D unimodular Lie groups. By a parameter count one can check that all such control systems are equivalent to a finite number of model ones. After the classification is done, minimum controllability time will be estimated. This problem has some applications in quantum control, where the minimum controllability time is called quantum speed limit.

1 Introduction

As motivation, the Schrodinger equation is

$$i\dot{\psi} = -\frac{\Delta}{2}\psi + \sum u_i(t)V_i(t)$$

In this project, we are studying the control system

$$\dot{g} = (A + uB)g, \quad G \subset GL(n) \tag{1.1}$$

where u is the control input. We are particularly interested in working with the case $dimG = 3$, the unimodular Lie groups of dimension 3, which can be completely classified into five types of Lie groups: $SL(2)$, $SU(2)$, H^3, $SE(2)$, and $SH(2)$.

This project involves two key goals

1. Classify systems of the type 1.1 on all unimodular Lie groups of dimension 3
2. Find all examples of this system 1.1 which are controllable in arbitrary small time

Todo: need to give a lot more motivation.

2 Preliminaries

2.1 Groups

We first give the basic notions of groups, along with some examples.

Definition 2.1. *A group G is a non-empty set with an operation $G \times G \to G$, $(a, b) \mapsto ab$ such that the following axioms hold:*

1. $(ab)c = a(bc)$ *(associativity)*
2. $\exists e \in G$ *such that* $eg = ge$ *for all* $g \in G$ *(identity)*
3. $\forall g \in G$, $\exists d \in G$ *such that* $dg = gd = e$ *(inverse)*

A group is called commutative or Abelian if for all $g, h \in G$, we have $gh = hg$.

Example 2.2. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}$, $M_2(\mathbb{F})$, $\mathbb{Z}/n\mathbb{Z}$, *all with the addition operation, are groups.*

Example 2.3. *Some more examples of groups*

- *Permutation group* S_n
- *Dihedral group* D_8
- *General linear group* $GL(n, \mathbb{F})$

Definition 2.4. *Let G be a group. A subset $H \subset G$ is called a subgroup if*

1. $e \in H$
2. $a, b \in H \Rightarrow ab \in H$
3. $a \in H \Rightarrow a^{-1} \in H$

A subgroup H is a group in its own right.

Definition 2.5. *Let G be a group. G is called cyclic if there is a $g \in G$ such that any element of G is a power of g, i.e. $G = \{g^n : n \in \mathbb{Z}\}$. This element g is called a generator of G.*

Next, we introduce group actions on sets.

Definition 2.6. *Let G be a group and let S be a non-empty set. G acts on S if we are given a function $G \times S \to S$, $(g, s) \mapsto gs$ such that*

1. $es = s$, $\forall s \in S$
2. $(g_1 g_2)s = g_1(g_2 s)$, $\forall g_1, g_2 \in G, s \in S$

Definition 2.7. *We are given an action of G on S, and an element $s \in S$ The orbit of s is*

$$Orb(s) = \{gs : g \in G\}$$

The stabiliser of s is

$$Stab(s) = \{g \in G : gs = s\}$$

Note that $Orb(s)$ is a subset of S, while $Stab(s)$ is a subset (in fact a subgroup) of G. Below are some examples.

Example 2.8. *Let the group be $G = S_n$, let the set be $T = \{1, 2, \ldots, n\}$. Then for $i \in T$, $\sigma i = \sigma(i)$, and $Orb(i) = T$.*

Example 2.9. *Let the group be $G = \mathbb{R}$, let the set be $S =$ sphere in $\mathbb{R}^3$ of radius 1 about the origin. $\mathbb{R}$ acts by rotating the sphere around the z-axis. The element $r \in \mathbb{R}$ rotates r radians. For $s \in S$, $Orb(s)$ is the altitude line on the sphere.*

Example 2.10. *Let G be a group, let $H \subset H$ be a subgroup. Here, we let H be the group and G be the set. H acts on G by $H \times G \to G$, $(h, g) \mapsto hg$. For $g \in G$, $Orb(g) = \{hg : h \in H\} = Hg =$ right cosets of H.*

Definition 2.11. *A group action of G on X is called transitive if $\forall x, y \in S$, $\exists g \in G$ such that $gx = y$.*

2.2 The Exponential Map

Let $M_n(\mathbb{F})$ be the space of $n \times n$ matrices with coefficients in the field $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). We consider it as a normed vector space with norm

$$||X|| = \sup_{u \neq 0} \frac{|X(u)|}{|u|}$$

where $|u|$ is the usual norm in $\mathbb{R}^n$ or the hermitian norm in $\mathbb{C}^n$.

Definition 2.12. *For $X \in M_n(\mathbb{F})$, we define*

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

Using the Cauchy-Schwartz inequality and the property $||XY|| \leq ||X||||Y||$, we can conclude that this series is normally convergent on compact sets.

Theorem 2.13. *Let $X \in M_n(\mathbb{R})$ be a fixed matrix. Then the map $\exp : \mathbb{R} \to M_n(\mathbb{R})$ defined by*

$$t \to \exp(tX)$$

is analytic and is the unique solution to the differential equation

$$\frac{dY}{dt} = XY$$

with initial condition $Y(0) = Id$.

Proposition 2.14. *The exponential map satisfies the following properties*

1. *For any $g \in GL(n, \mathbb{R})$ and $X \in M_n(\mathbb{R})$,*

$$g(\exp(X))g^{-1} = \exp(gXg^{-1})$$

2. *If $X, Y \in M_n(\mathbb{R})$ are two commuting matrices then*

$$\exp(X + Y) = \exp(X)\exp(Y)$$

3. *For any $X \in M_n(\mathbb{R})$*

$$\exp(-X) = (\exp(X))^{-1}$$

Proposition 2.15 (Jacobi's formula)**.** *For any $X \in M_n(\mathbb{F})$ we have*

$$\det(\exp X) = e^{tr(X)}$$

2.3 Lie Groups, Lie Algebras

Definition 2.16. *A Lie group is a topological group which is a smooth manifold and such that the group operations, product $G \times G \to G$ and inverse map $G \to G$*

$$(g, h) \mapsto gh \text{ and } g \mapsto g^{-1}$$

are smooth.

Proposition 2.17. *The group of invertible matrices of rank n, $GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$, is a Lie group.*

We will mostly be dealing with Lie groups which are submanifolds of $GL(n, \mathbb{R})$, which are called linear Lie groups.

Definition 2.18. *A Lie algebra over a field $\mathbb{F}$ is a vector space $\mathfrak{g}$ over $\mathbb{F}$ equipped with a bilinear map (called a Lie bracket)*

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

such that

1. *$[X, Y] = -[Y, X]$ (skew-symmetry)*
2. *$[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)*

Definition 2.19. *A Lie subalgebra of $\mathfrak{g}$ is a vector subspace $\mathfrak{h}$ such that for all $X, Y \in \mathfrak{h}$, $[X, Y] \in \mathfrak{h}$.*

Example 2.20. *The vector space $M_n(\mathbb{F})$ is a Lie algebra with the Lie bracket defined for all $X, Y \in M_n(\mathbb{F})$ by $[X, Y] = XY - YX$.*

To any Lie group, we can associate a Lie algebra. It is the tangent space at the identity element of the group. First, in the case of linear Lie groups, we can develop this idea without knowledge of the tangent space to a manifold.

Definition 2.21. *Let $G \subset GL(n, \mathbb{R})$ be a linear group. Define*

$$Lie(G) = \{X \in M_n(\mathbb{R}) : \exp(tX) \in G \ \forall t \in \mathbb{R}\}$$

Proposition 2.22. *Let $G \subset GL(n, \mathbb{R})$ be a linear group. Then $Lie(G) \subset M_n(\mathbb{R})$ is a Lie subalgebra.*

Definition 2.23. *Let G be a linear Lie group and let $\mathfrak{g}$ be its Lie algebra.*

1. *The adjoint representation of the Lie group is the homomorphism*

   $$Ad : G \to Aut(\mathfrak{g})$$

   given by $Ad(g) = Ad_g$.

2. *The adjoint representation of the Lie algebra is the homomorphism*

   $$ad : \mathfrak{g} \to End(\mathfrak{g})$$

   given by $ad(X) = ad_X$.

We refer to Ad_g as the adjoint map defined by g, and to Ad as the adjoint map. We say the derivation ad_X is the adjoint map defined by X. The following gives a relation between these maps.

Proposition 2.24. *Let G be a linear Lie group, let $\mathfrak{g}$ be its Lie algebra, and $X \in \mathfrak{g}$. Then*

$$\frac{d}{dt} Ad_{\exp(tX)|_{t=0}} = ad_X$$

2.4 Examples of Lie Groups, Lie Algebras

In the following two examples, $\mathbb{R}$ can be replaced by $\mathbb{C}$.

Example 2.25. *Some important Lie groups*

1. $GL(n,\mathbb{R}) = \{g \in M_n(\mathbb{R}) : \det g \neq 0\}$
2. $SL(n,\mathbb{R}) = \{g \in GL(n,\mathbb{R}) : \det g = 1\}$
3. $O(n,\mathbb{R}) = \{g \in GL(n,\mathbb{R}) : {}^t gg = I_n\}$
4. $SO(n,\mathbb{R}) = O(n,\mathbb{R}) \cap SL(n,\mathbb{R})$
5. $U(n) = \{g \in GL(n,\mathbb{C}) : {}^t\bar{g}g = I_n\}$
6. $SU(n) = U(n) \cap SL(n,\mathbb{C})$

Example 2.26. *Some important Lie algebras*

1. $\mathfrak{gl}(V)$ *and* $\mathfrak{gl}(n,K)$*, respectively corresponding to* $End_{\mathbb{K}}(V)$ *and* $M_n(K)$*, where* K *is a field and* V *is a vector space over* K
2. $\mathfrak{sl}(n,\mathbb{R}) = \{X \in \mathfrak{gl}(n,\mathbb{R}) : tr(X) = 0\}$
3. $\mathfrak{so}(n,\mathbb{R}) = \{X \in \mathfrak{gl}(n,\mathbb{R}) : {}^tX + X = 0\}$
4. $\mathfrak{u}(n,\mathbb{R}) = \{X \in \mathfrak{gl}(n,\mathbb{C}) : {}^t\bar{X} + X = 0\}$
5. $\mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n,\mathbb{C})$

Todo: need a more complete list, with classical Lie groups and algebras.

In particular for this project, we are focusing on the Lie groups $SL(2)$, $SU(2)$, H^3, $SE(2)$, and $SH(2)$.

1. $SL(2,K) = \{X \in M_2(K) : \det A = 1\}$.
2. $SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}$.
3. $H_3(A) = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in A \right\}$, where A is a unitary ring.
4. $SE(n)$ includes arbitrary combinations of translations and rotations in Euclidean space, but not reflections (to be distinguished with $E(n)$). These transformations preserve Euclidean distance.
5. $SH(2)$ is the same as $SE(n)$, except "rotation" is replaced by "hyperbolic rotation". These transformations preserve the Lorentz form.

2.5 Killing Form

A property of finite dimensional Lie algebras is that they come with a symmetric bilinear form, which will be important in the structure theory of Lie algebras.

Definition 2.27. *Let* $\mathfrak{g}$ *be a finite dimensional Lie algebra. The Killing form of* $\mathfrak{g}$ *is the bilinear form* B *on* $\mathfrak{g}$ *defined by*

$$B(X,Y) = tr(ad(X) \circ ad(Y)), \quad \forall X, Y \in \mathfrak{g}$$

Proposition 2.28. *Let $\mathfrak{g}$ be a finite dimensional Lie algebra, and B its Killing form. Then B is symmetric and ad-invariant, i.e.*

$$B(ad(X)Y, Z) + B(Y, ad(X)Z) = 0, \quad \forall X, Y, Z \in \mathfrak{g}$$

Perhaps a clearer way to express the symmetry property is to write $B(X, Y) = B(Y, X)$. Moreover, ad-invariance means $ad(\mathfrak{g})$ is included in the Lie algebra $\mathfrak{o}(B)$ of the Lie group $O(B)$, which is the group of linear isomorphisms of $\mathfrak{g}$ preserving B.

Proposition 2.29. *Let G be a Lie group with Lie algebra $\mathfrak{g}$. For all $g \in G$, the adjoint map $Ad(g)$ preserves the Killing form of $\mathfrak{g}$, i.e.*

$$B(Ad(g)X, Ad(g)Y) = B(X, Y), \quad \forall X, Y \in \mathfrak{g}$$

The proof of this proposition makes use of the property that

$$ad(Ad(g)X) = Ad(g) \circ ad(x) \circ Ad(g)^{-1}$$

as well as the property that the trace is invariant by conjugation in $GL(\mathfrak{g})$.

It is useful to note that the adjoint map $ad(X) : \mathfrak{g} \to \mathfrak{g}$ is given by

$$ad(X)Y = [X, Y]$$

I also include the definition of the centre of a Lie group or Lie algebra.

Definition 2.30. *The centre of G is*

$$Z(G) = \{g \in G : \forall x \in G \; gx = xg\}$$

Definition 2.31. *The centre of $\mathfrak{g}$ is*

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g} \; [X, Y] = 0\}$$

Example 2.32. *Some examples of Killing forms:*

1. *The Killing form of $\mathfrak{gl}(n, \mathbb{R})$ is $B(X, Y) = 2ntr(XY) - 2tr(X)tr(Y)$.*
2. *The Killing form of $\mathfrak{sl}(n, \mathbb{R})$ is $B(X, Y) = 2ntr(XY)$.*
3. *The Killing form of an Abelian Lie algebra vanishes.*

Todo: add computational examples of finding the Killing form and the matrix of the Killing form.

2.6 Control Systems

Given two control systems, we can use the following theorem to characterize their equivalence.

Theorem 2.33. *Affine control systems are equivalent if and only if their associated affine distributions are diffeomorphic.*

Let G be a Lie group, and let $\mathfrak{g}$ be its associated Lie algebra. Consider the control system 1.1 with $g \in G$ and $A, B \in \mathfrak{g}$. When classifying 1.1 on this particular Lie group, we want to simplify the pair (A, B) into an equivalent pair, say (A', B'). This equivalency is denoted $(A, B) \sim (A', B')$. What tools can we use in this classification?

The first is the adjoint action of a Lie group on its Lie algebra, $ad : G \times \mathfrak{g} \to \mathfrak{g}$. For each $g \in G$, it is the derivative of this action in the second argument at the neutral element of G.

For a matrix Lie group G, it is simply the conjugation $ad_g X = gXg^{-1}$. After letting g act on both A and B, the Killing form $B(A, B)$ is invariant

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

Thus we can write the equivalency $(A, B) \sim (gAg^{-1}, gBg^{-1})$. In different Lie groups, the conjugation group action will have different interpretations, which will become clear in the following sections.

Second, we can rewrite the control u as $u = u' + \delta$ for any δ. This can be thought of as a perturbation to the control. Then,

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

and so we have the equivalence $(A, B) \sim ((A + \delta B), B)$.

Another way to modify the control is to scale the control by a constant, writing $u = \mu u'$. Then,

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

which shows the equivalence $(A, B) \sim (A, \mu B)$.

Finally, the fourth option is to perform time scaling, for example by choosing a new measurement of time. If we scale the time by a factor of $1/\lambda$, then there is the equivalence $(A, B) \sim (\lambda A, \lambda B)$. Clearly, $\lambda \neq 0$.

3 Classifying Control Systems on $SU(2)$

We consider the system 1.1 with $g \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. This is the most basic scenario because it is closely related to the usual Euclidean space $\mathbb{R}^3$.

The following matrices

$$u_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}; \quad u_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}; \quad u_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

form a basis for $\mathfrak{su}(2)$. The structural constants are given by the following relations

$$[u_1, u_2] = u_3; \quad [u_2, u_3] = u_1; \quad [u_3, u_1] = u_2$$

The factor of $1/2$ is useful in that the Killing form $B(u_i, u_j) = \delta_{ij}$ instead of $4\delta_{ij}$, where δ_{ij} is the Kronecker delta function.

4 Classifying Control Systems on $SL(2)$

We consider the system 1.1 with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. The following matrices

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form a basis for $\mathfrak{sl}(2)$. The structural constants are given by the following relations

$$[H, E] = 2E; \quad [H, F] = -2F; \quad [E, F] = H$$

This is the most commonly seen choice of basis, and it is the one I will use, for the most part. Note, however, that we could have alternatively chosen the basis

$$E = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

which can make the geometry of transformations more apparent.

5 Classifying Control Systems on H^3

We consider the system 1.1 with $g \in H^3$ and $A, B \in \mathfrak{h}$. The following matrices

$$X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{h}$. The structural constants are given by the following relations

$$[X, Y] = Z; \quad [X, Z] = 0; \quad [Y, Z] = 0$$

The Heisenberg group is different from the previous two scenarios in that for any $B \in \mathfrak{h}$, it can be verified that the Killing form $B(B, B) = 0$. So the Killing form is not so useful for classification here. Instead, we will have to use the centre $Z(H)$, as well as automorphism groups.

By simply applying the definition of the centre of a group, I found

$$Z(H) = \left\{ \begin{pmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : c \in \mathbb{R} \right\}$$